Supplementary Materials Accompanying "Title: The role of ecology in allopatric speciation of darters in the Central Highlands, USA"

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Data Availability:

- DNA sequence data is available at the NCBI archive - BioProject Accession Number PRJNA1177478.
- Meristic and morphological trait data and diet data are available in a Dryad repository (https://doi.org/10.5061/dryad.stqjq2cfv).
- Scripts are also available on the Dryad repository.

- Information about the tissues and museum specimens used in this study are provided in Supplementary Data Tables S1 and S2.

Supplementary Methods

Our strategy was to sample at every stream in the range of each species at historical collection sites and localities identified using Google Earth with abundant rocky habitat. Collection methods included snorkeling to sight individuals which were then captured with hand nets while underwater, backpack electrofishing, and occasionally seining. Specimens were anesthetized using MS222 and a portion of the right pectoral fin was removed and preserved in 95-100% ethanol for subsequent DNA isolation. Collected specimens were fixed in 10% formalin for one to two weeks, rinsed in water, and transferred to 70% ethanol for long-term preservation in the research collection of fishes in the Division of Vertebrate Zoology at the Yale Peabody Museum. This study was performed under the auspices of Yale University IACUC (No. 2015-10681; 2018-10681; 2021-10681).

DNA was extracted from tissue samples using DNeasy extraction kits following the manufacturer's protocol (QIAGEN) in Yale University lab facilities. DNA concentrations were quantified using a Qubit Fluoremeter (Thermo Fisher Scientific) and quality was assessed with gel electrophoresis. We used 20 µL of ~400 ng genomic DNA for library preparation. DNA extractions were digested using PstI/MspI restriction enzymes, ligated with 96 unique barcodes, and amplified via polymerase chain reaction (PCR). Following PCR, samples were normalized for DNA concentrations and cleaned using a QIAquick PCR purification kit (QIAGEN) per the manufacturer's protocol. Each plate of 96 samples was pooled into one library, size selected for fragments between 300 and 500 bp and sequenced using 100 bp single-end sequencing on an Illumina Hiseq 4000 at the University of Oregon GC3F facility (https://gc3f.uoregon.edu).

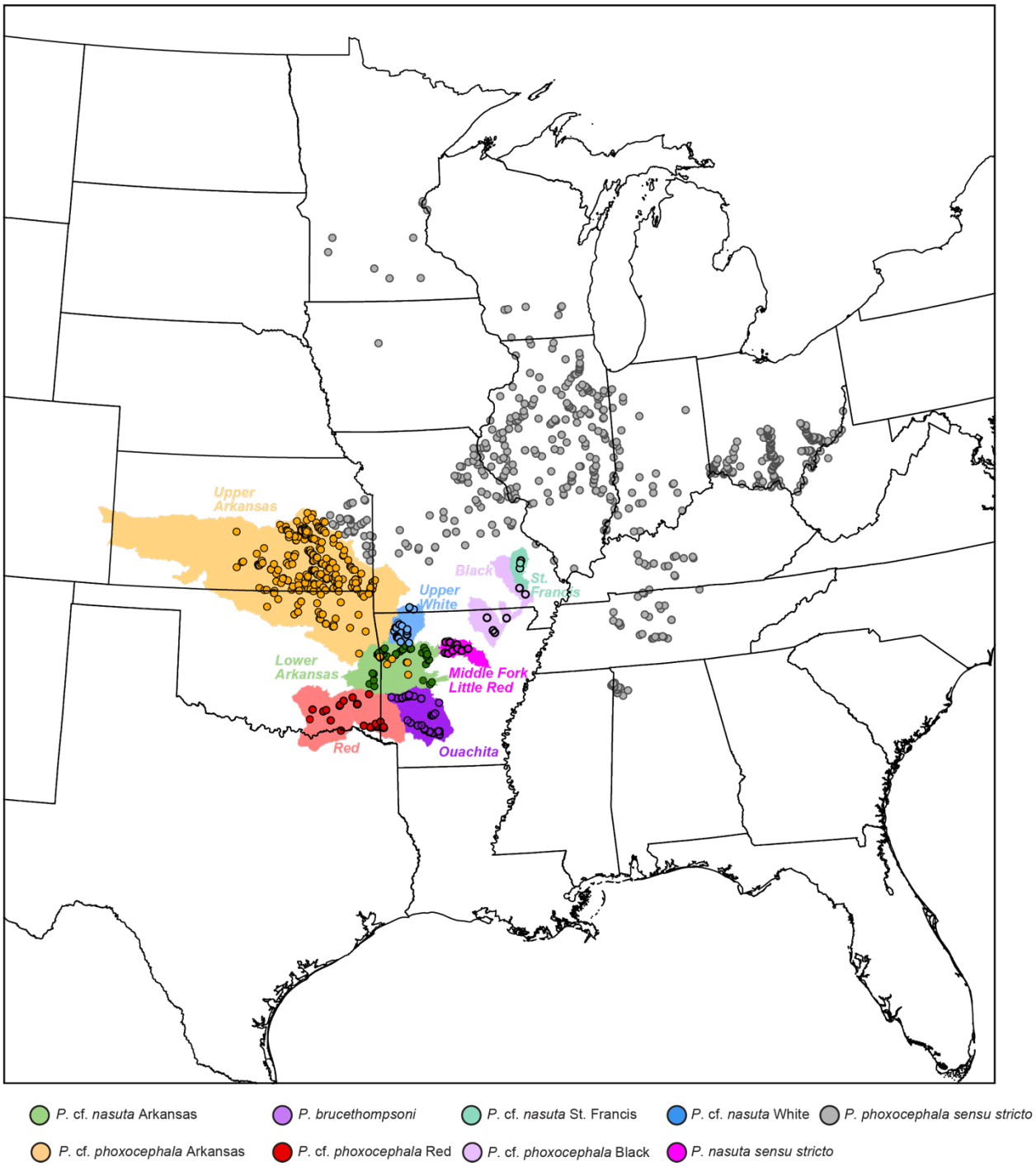

Figure S1: Full geographic distribution of *Percina phoxocephala, P. nasuta,* and *P. brucethompsoni. P. phoxocephala sensu stricto* is not included in our ecological analyses. The points are museum collection localities compiled from FishNet2 (http://www.fishnet2.net/). The shaded background represents the river systems boundaries associated with the study species (excluding *Percina phoxocephala sensu stricto*). *P.* cf. *phoxocephala* Arkansas (yellow) is distributed mostly in the upper Arkansas River system, but it is also present in the lower Arkansas River system where it may co-occur with *P.* cf. *nasuta* Arkansas (green).

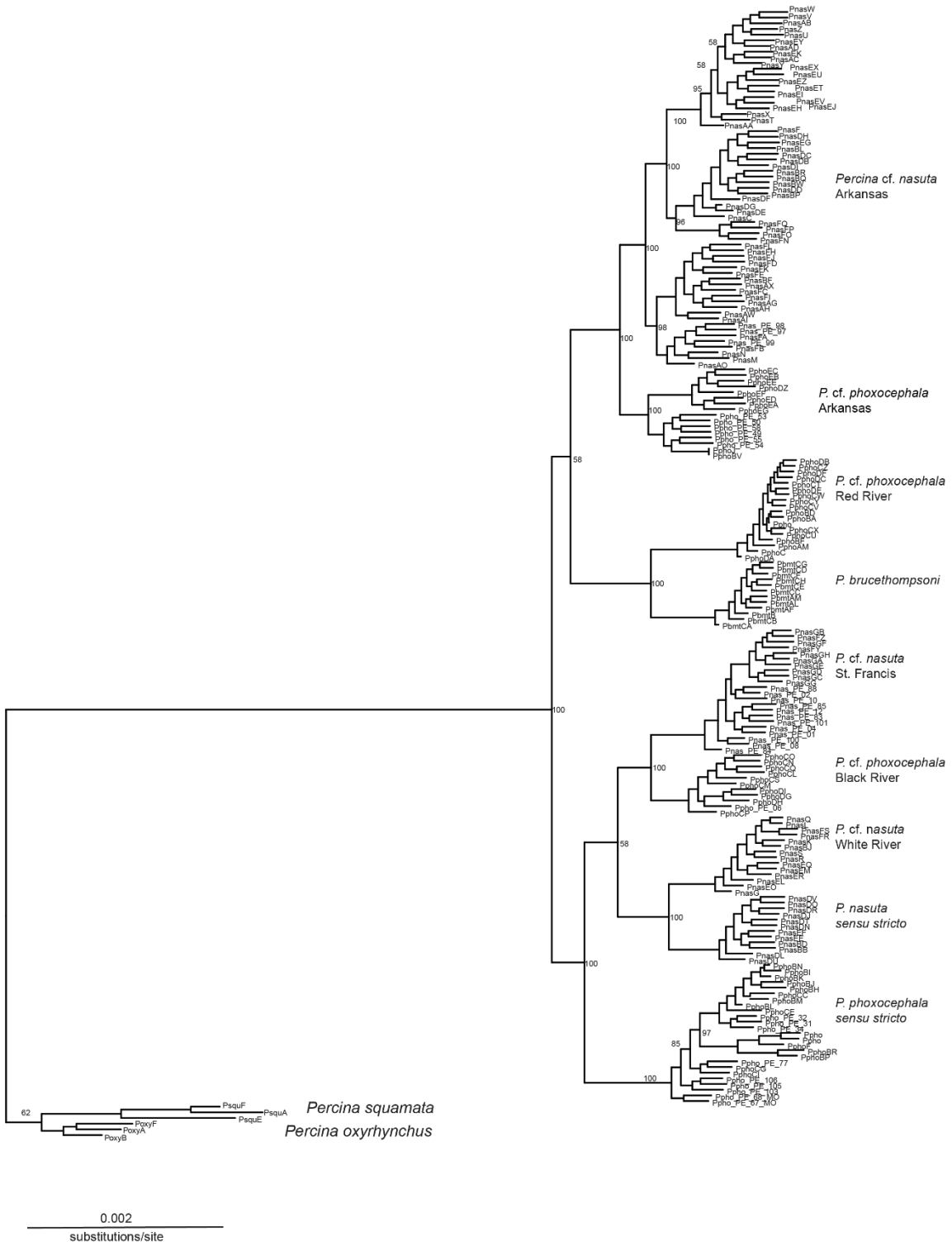

Fig. S2: Maximum likelihood phylogeny inferred from concatenated double-digest restriction site associated DNA (ddRAD) sequence data for the study group of *Percina* species in the Central Highlands, USA. The figure is identical to Figure 1 in the main text, but here the outgroups and clone codes that correspond to individual specimens are included. Additional specimen information can be acquired in Data Table S1.

Table S1: G-PhoCS estimates of the number of migrants per generation (*Nm*) among sister species pairs and 95% credible interval (CI)

Species pair	*Nm*	95% CI
From *Percina* cf. *nasuta* Arkansas to *P*. cf. *phoxocephala* Arkansas	0.34	0.08, 0.69
From *Percina* cf. *phoxocephala* Arkansas to *P*. cf. *nasuta* Arkansas	0.39	0.14, 0.87
From *Percina* cf. *nasuta* St. Francis to *P*. cf. *phoxocephala* Black	8.57×10^{-6}	0.0, 1.44×10^{-4}
From *Percina* cf. *phoxocephala* Black to *P*. cf. *nasuta* St. Francis	5.82×10^{-6}	0.0, 9.57×10^{-6}
From *Percina brucethompsoni* to *P*. cf. *phoxocephala* Red	4.60×10^{-5}	2.31×10^{-9}, 2.26×10^{-4}
From *Percina phoxocephala* Red to *P. brucethompsoni*	2.50×10^{-5}	1.73×10^{-9}, 1.27×10^{-4}
From *Percina* cf. *nasuta* White to *P. nasuta*	6.48×10^{-6}	2.86×10^{-7}, 2.18×10^{-5}
From *Percina nasuta* to *P*. cf. *nasuta* White	0.34	0.12, 0.74

Table S2: Estimates of the genealogical divergence index (*gdi*) using different numbers of loci for the eight lineages investigated in the study as well as *Percina phoxocephala sensu stricto* (mean ± 2σ). Intermediate values (>0.2) are consistent with either species-level or population-level divergence. High values (> 0.7) are consistent with species-level divergence. The estimate presented in the main text (Table 1) uses 1,729 loci. The mean +/- two standard deviations is shown.

Number of Loci	*P. brucethompsoni*	*P.* cf. *phoxocephala* Red	*P.* cf. *nasuta* Arkansas	*P.* cf. *phoxocephala* Arkansas	*P.* cf. *nasuta* St. Francis	*P.* cf. *phoxocephala* Black	*P.*cf. *nasuta* White	*P. nasuta*	*P. phoxocephala*
250	0.72 ± 0.08	0.80 ± 0.07	0.36 ± 0.21	0.33 ± 0.13	0.44 ± 0.11	0.41 ± 0.14	0.58 ± 0.08	0.46 ± 0.17	1.00 ± 0.02
1729	0.76 ± 0.03	0.85 ± 0.02	0.28 ± 0.19	0.34 ± 0.04	0.55 ± 0.04	0.40 ± 0.05	0.57 ± 0.03	0.45 ± 0.13	1.0 ± 0.0003
7302	0.80 ± 0.02	0.85 ± 0.02	0.40 ± 0.12	0.38 ± 0.03	0.58 ± 0.04	0.46 ± 0.04	0.60 ± 0.03	0.51 ± 0.06	0.97 ± 0.05

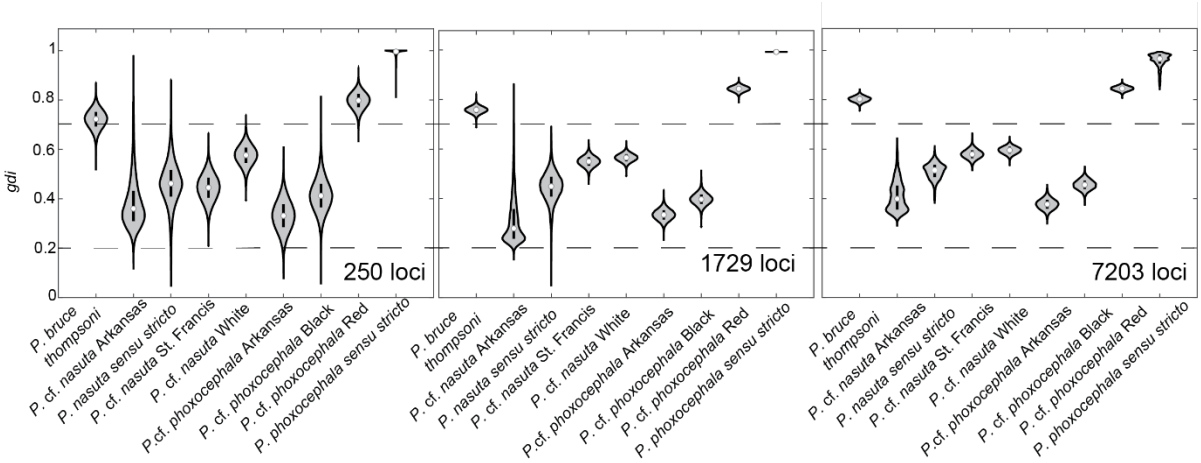

Figure S3: Violin plots showing the distribution of estimates of the genealogical divergence index (*gdi*) using different numbers of loci for the eight lineages investigated in the study in addition to *Percina phoxocephala sensu stricto* (mean ± 2σ). The results in the main text (Table 2) use 1729 loci. Values below 0.2 are consistent with population-level divergence. Values over 0.7 are consistent with species-level divergence and intermediate values can be consistent with distinct species if supported by phenotypic disparity.

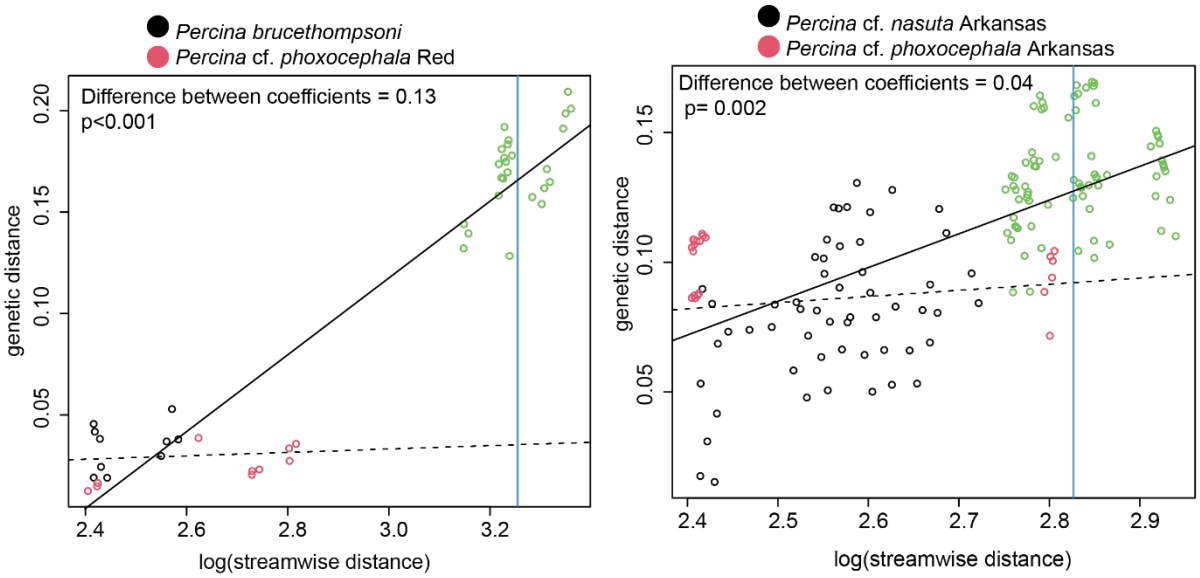

Figure S4: Testing species delimitation using the relationship between genetic and geographic distance following the methods of Hausdorf and Henning (2020). The black and red points represent the within-species pairwise comparisons for putative *Percina* species in the Central Highlands, USA. The green points represent the between-species comparisons. First, we tested the null hypothesis that the within-species comparisons can be modelled by a single regression. In both analyses presented here, the null hypothesis was not rejected ($p > 0.05$). Thus, we use one regression for the 'within-species' comparison (dotted black line fit to black and red points) and one 'between-species' regression (solid black line fit to all data). For both putative sister species pairs analyzed here, the coefficients for the within-species regression are less than the between-species regression ($p < 0.05$). In other words, the within-species relationship between geographic and genetic distance is not sufficient to explain the degree of genetic distance present at the geographic center of the between-group distances (blue line). The result supports delimitation.

The test is not performed for the other putative sister species pairs because there were too few localities with sufficient geographic distance from each other to perform within-species regressions.

	Actual								
Predicted	*P. nasuta sensu stricto*	*P.* cf. *nasuta* White	*P.* cf. *nasuta* Arkansas	*P.* cf. *phoxocephala* Arkansas	*P. brucethompsoni*	*P.* cf. *phoxocephala* Red	*P.* cf. *nasuta* St. Francis	*P.* cf. *phoxocephala* Black	*P. phoxocephala* sensu stricto
P. nasuta sensu stricto	25	2	19	0	5	0	0	0	0
P. cf. *nasuta* White	2	16	8	0	0	0	1	1	0
P. cf. *nasuta* Arkansas	43	38	136	7	10	3	4	3	1
P. cf. *phoxocephala* Arkansas	0	5	9	116	0	10	3	6	25
P. brucethompsoni	8	2	5	0	35	0	0	0	0
P. cf. *phoxocephala* Red	0	0	0	6	0	34	10	4	5
P. cf. *nasuta* St. Francis	0	2	1	1	0	2	12	1	2
P. cf. *phoxocephala* Black	0	0	1	2	0	0	0	3	0
P. phoxocephala sensu stricto	0	0	1	18	0	7	2	0	78
Percent Positively Identified	**32 %**	**25 %**	**76 %**	**78 %**	**70 %**	**61 %**	**38 %**	**17 %**	**70 %**

Table S3: A confusion matrix showing the results from a linear discriminant analysis (LDA) of meristic traits. The bottom row shows the percent of the specimens positively identified as the correct species. Note that sister species are more effectively distinguished from each other than from the entire dataset (e.g. only nine specimens of *Percina* cf. *nasuta* Arkansas are incorrectly identified as its sister *P.* cf. *phoxocephala* Arkansas (5%), but 19 are identified as *P. nasuta sensu stricto* (11%)).

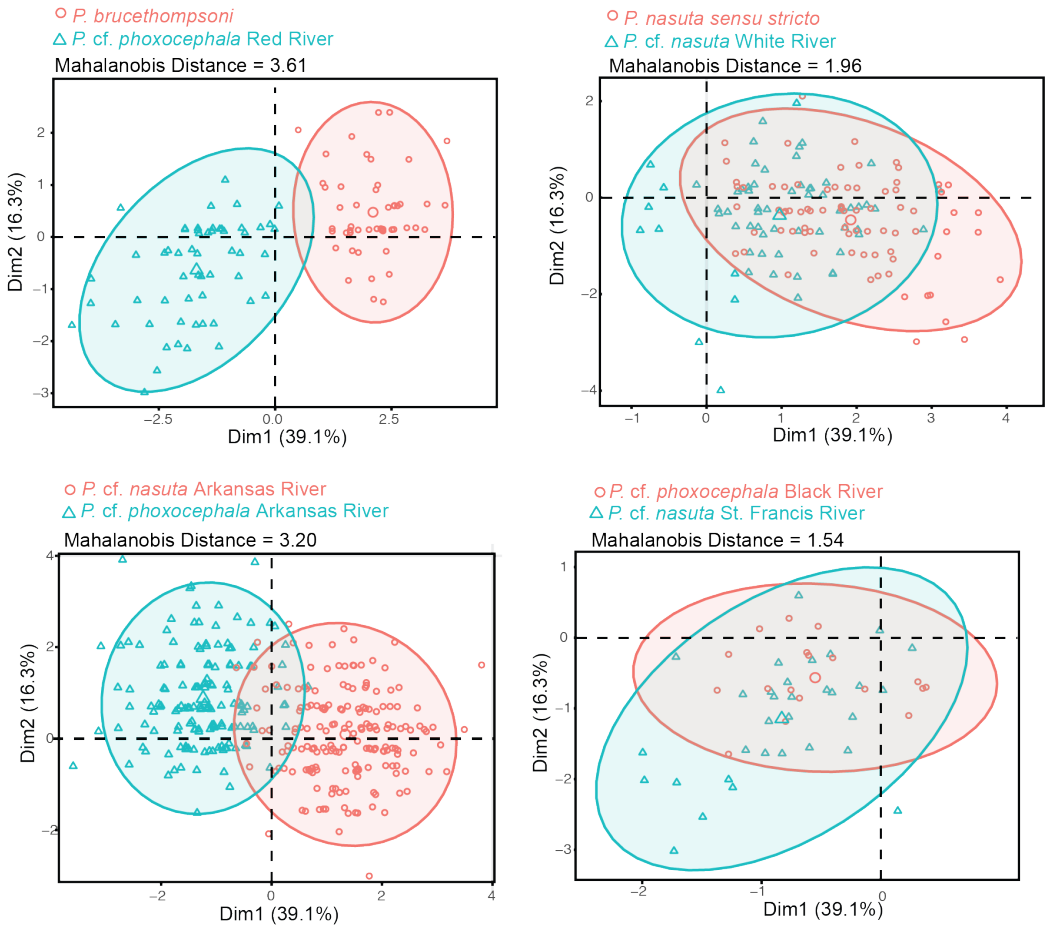

Figure S5: Principal component analysis of meristic trait data for *Percina* species. The pairwise Mahalanobis distance between each of the comparisons between sister species is reported on top of each panel.

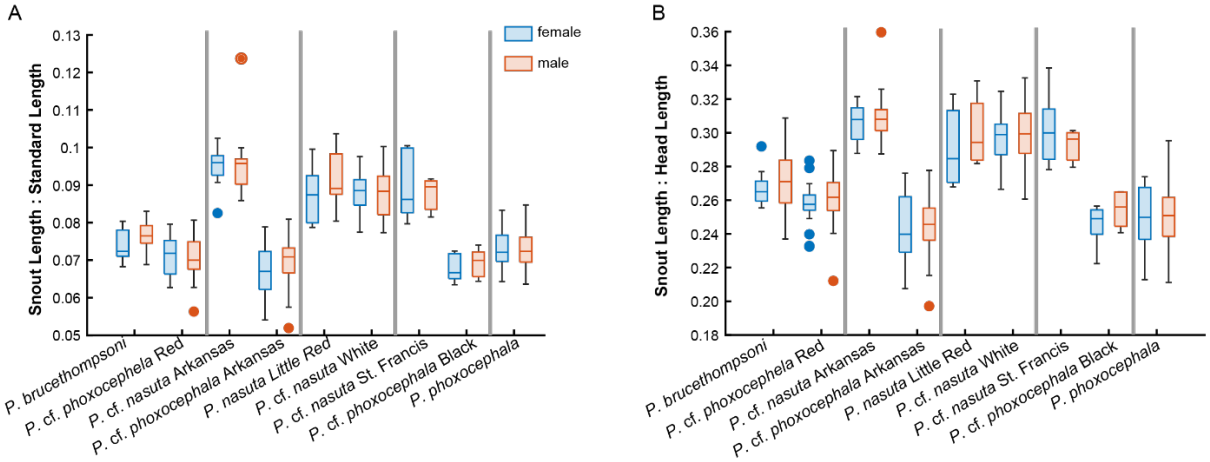

Figure S6: Box-and-whisker plot showing the **A)** ratio of snout to standard length and **B)** ratio of snout to head length, grouped by sex for delimited *Percina* species in the Central Highlands, USA. Two-tailed Wilcoxon rank-sum tests indicate no significant difference between median values in snout length ratios between the sexes within any lineage ($p<0.05$). Figure 2 in the main text includes the same data as **B)**, however, in the main text, the data are not separated by sex. Additionally, *P. phoxocephala sensu stricto* is shown here, and not the main text.

Variable	Category	Description and Source	Significant Loading
Elevation	Topography	NHD Flowline (NHD)	PC1
Slope	Topography Hydrology	NHD Flowline (NHD)	PC2
Forest	Landcover	% Catchment (NLCD)	PC1
Maximum Annual Temperature	Climate	Catchment-Averaged (EPA)	PC1
Mean Annual Precipitation	Climate	Catchment-Averaged (EPA)	PC1
Mean Annual Discharge	Hydrology	NHD Flowline (NHD)	PC2, PC7
Depth to Bedrock	Alluvial Substrate	Catchment-Averaged (EPA)	PC2
Sand	Alluvial Substrate	% Catchment (EPA)	PC4
Detrital Sediment	Alluvial Substrate	% Catchment (SGMC)	PC2
Clastic Bedrock	Bedrock Substrate	% Catchment (SGMC)	PC7
Shale Bedrock	Bedrock Substrate	% Catchment (SGMC)	PC3
Igneous Bedrock	Bedrock Substrate	% Catchment (SGMC)	PC6
Chert Bedrock	Bedrock Substrate	% Catchment (SGMC)	PC5, PC7
Carbonate Bedrock	Bedrock Substrate	% Catchment (SGMC)	PC3, PC4

Table S4: Variables included in principal component analysis. NHD = National Hydrography Dataset (https://www.epa.gov/waterdata/nhdplus-national-hydrography-dataset-plus). NLCD = National Land Cover Dataset (https://www.usgs.gov/centers/eros/science/national-land-cover-database). EPA = Environmental Protection Agency StreamCat Dataset (Hill et al., 2016). SGMC = State Geologic Map Compilation (Horton et al., 2017).

Species Pairs	Degrees of Freedom	F	R^2	p-value
Percina cf. *nasuta* Arkansas *Percina* cf. *phoxocephala* Arkansas	(1,43)	28	0.40	0.004
Percina brucethompsoni *Percina* cf. *phoxocephala* Red	(1,44)	10	0.19	0.004
Percina cf. *nasuta* St. Francis *Percina* cf. *phoxocephala* Black	(1,30)	2	0.05	0.338
Percina cf. *nasuta* White *Percina nasuta sensu stricto*	(1,30)	1.6	0.05	0.338

Table S5: Results of PERMANOVA to test for differences in gut contents between sister species. In the main text, the analysis includes season of collection, here it does not. Degrees of freedom are reported for the predictor and the residual (predictor, residual). P-values are adjusted for multiple comparisons using a Holm-Bonferroni correction.

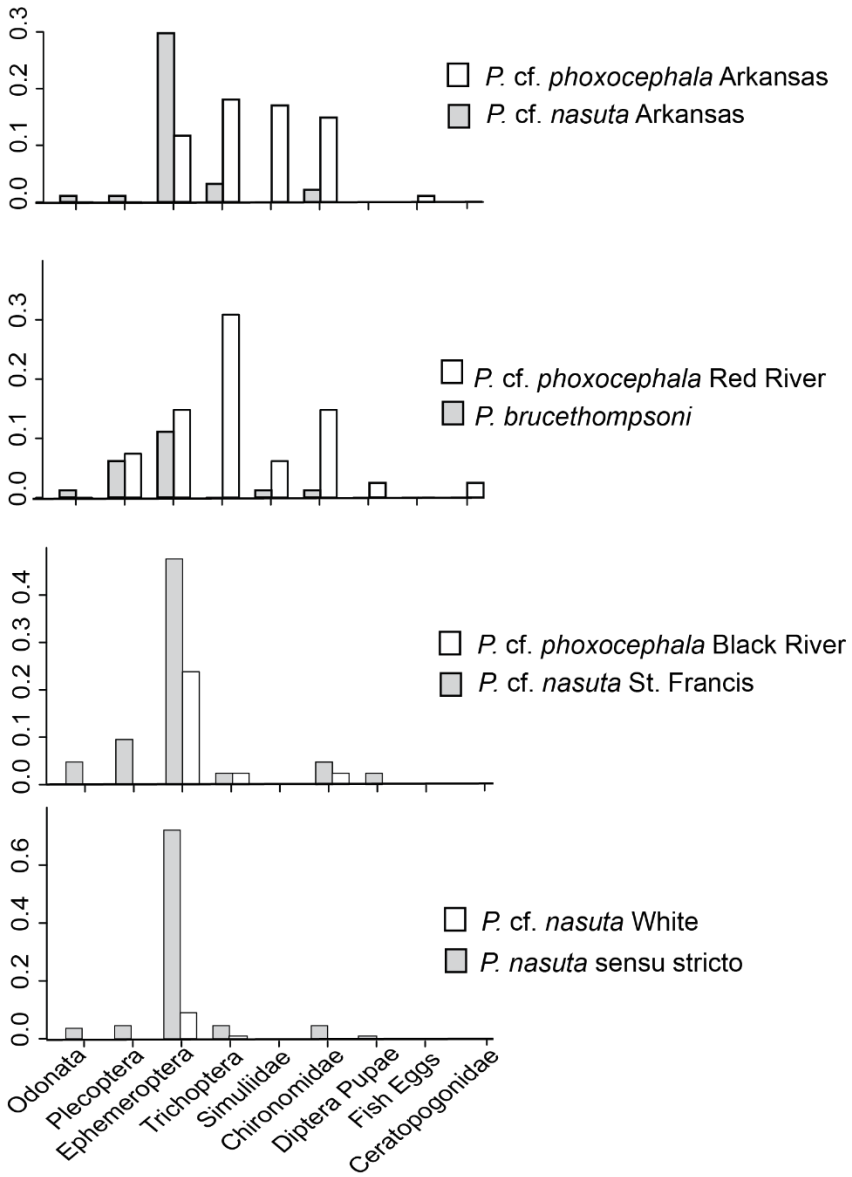

Figure S7: Relative frequency of stomach contents analyzed in museum specimens of *Percina* species included in this study and identified to the order or family level.

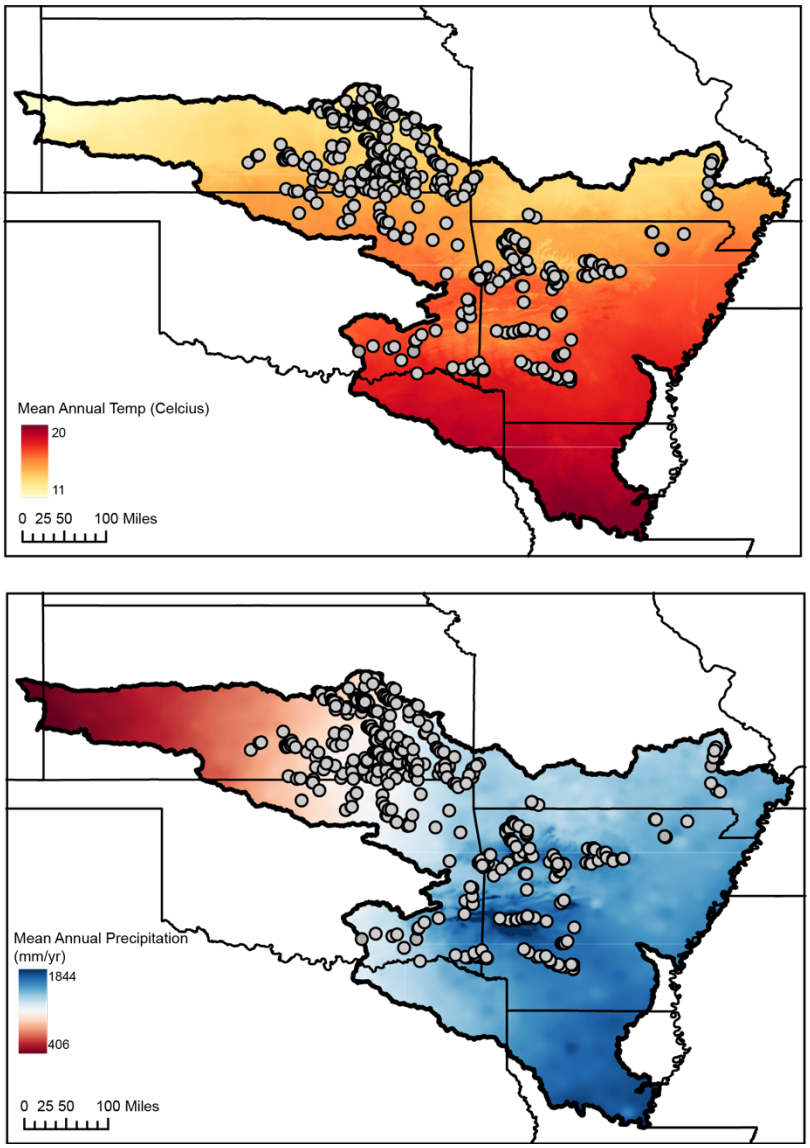

Fig. S8: Mean annual temperature and precipitation from across the range of the study species (darters) in the Central Highlands, USA. Localities are shown with grey points. Climate data from PRISM (https://prism.oregonstate.edu/).

Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

References Cited in Supplement

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Horton, J. D., San Juan, C. A., & Stoeser, D. B. (2017). *The State Geologic Map Compilation (SGMC) geodatabase of the conterminous United States*, No 1052.